

MiniProject2_WebScraping

Cynthia Budiyanto and Willemijn Leiner

```
library(rvest)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter()      masks stats::filter()
x readr::guess_encoding() masks rvest::guess_encoding()
x dplyr::lag()          masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(stringr)
library(polite)
library(htmltools)
library(purrr)
```

```
# Defining the url to be its own value to make it easier to work with
url <- "https://link.springer.com/search?facet-journal-id=12968&query=machine+learning&content-type=article"

# Parse the HTML content of the page
page <- read_html(url)

# Extracting article titles using the CSS selector
titles <- page |>
  html_elements("a.app-card-open__link") |>
  html_text()

titles
```

[1] "\n	Precision measurement of cardiac structure and function
[2] "\n	A deep learning approach for fully automated cardiac MRI segmentation
[3] "\n	Radiomics and deep learning for myocardial strain estimation
[4] "\n	Highlights of the Virtual Society for Cardiovascular Medicine
[5] "\n	Deep learning to diagnose cardiac amyloidosis
[6] "\n	Machine learning in cardiovascular magnetic resonance
[7] "\n	Automatic segmentation of the great arteries
[8] "\n	Accelerated cardiac T1 mapping in four heartbeats
based T1 estimation approach\n	\n
[9] "\n	Fully-automated global and segmental strain estimation
[10] "\n	Machine learning derived segmentation of phase-contrast MRI
[11] "\n	Training and clinical testing of artificial neural networks
[12] "\n	Fully-automated deep-learning segmentation of cardiac MRI
[13] "\n	Assessment of right ventricular size and function
[14] "\n	4D Flow cardiovascular magnetic resonance correlation
[15] "\n	Localized strain characterization of cardiomyopathy
[16] "\n	Le Cœur en Sabot: shape associations with advanced
[17] "\n	2021-2022 state of our JCMR\n
[18] "\n	Technical development and feasibility of a reproducible
[19] "\n	Interventional cardiovascular magnetic resonance
of-the-art\n	\n
[20] "\n	Quantitative susceptibility mapping (QSM) of myocardium

```
# Extracting relative URLs for each article
links <- page |>
  html_elements("a.app-card-open__link") |>
  html_attr("href")
links <- paste0("https://link.springer.com", links)

# Extracting publication date
dates <- page |>
  html_elements(".app-card-open__meta") |>
  html_text()

# Cleaning the date strings
dates_clean <- dates |>
  str_squish()
dates_clean <- dates_clean |>
  str_remove("Open access")
dates_clean <- dates_clean |>
  str_squish()
dates_clean
```

```
[1] "10 March 2022"      "27 February 2023" "27 June 2022"      "20 June 2022"
[5] "07 December 2020"  "07 October 2019"  "07 November 2022"  "06 January 2022"
[9] "11 March 2021"      "07 January 2019"  "07 April 2022"      "30 November 2020"
[13] "11 April 2022"      "20 July 2023"      "16 February 2023"  "04 August 2022"
[17] "31 December 2022"  "04 December 2023" "14 August 2023"     "18 August 2022"
```

```
# Combining the extracted data into a single tibble for inspection
articles <- tibble(
  title = titles,
  date = dates_clean,
  link = links
)

view(articles)
```

FUNCTION 1: SCRAPING INDIVIDUAL ARTICLE LINKS

```
get_link_from_search <- function(page, css_code) {
  link <- page |>
    html_elements(css_code) |>
    html_attr("href")
}
# defining the function
# piping the object 'page'
# find all html elements
# extracts the value of href

scrape_list_links1 <- function(url) {
  Sys.sleep(3)
  session <- bow(url, force = TRUE)
  page <- scrape(session)
  list_links <- get_link_from_search(page, ".app-card-open__link")
  complete_link <- paste0("https://link.springer.com", list_links)
  list(complete_link)
}
# defining the function
# polite scraping
# creates a browser session
# scrape the given URL
# extract the links using the function
# stitching the extracted links with the base URL
# wraps it into a list
```

FUNCTION 2: USING LINKS TO GET INDIVIDUAL ARTICLE INFO

```
get_info_from_page <- function(page, css_code) {
  result <- page |>
    html_elements(css_code) |>
    html_text(trim = TRUE)
}
# defining the function
# piping the object 'page'
# find all of the html elements
# retrieve text content
```

```

if (length(result) == 0) # if length of the s
  return(NA_character_) # then return the va
result[1] # take the first mat
}

scrape_individual_article <- function(x) { # defining the funct
  Sys.sleep(3) # polite scraping
  session <- bow(x, force = TRUE) # creates a browser s
  page <- scrape(session) # scrape the given URL
  jcmr_title <- get_info_from_page(page, # scrape the "title"
                                ".c-article-title")
  jcmr_vol <- get_info_from_page(page, # scrape the "volume"
                                ".c-article-identifiers--cite-list .c-article-identifiers__item:1")
  jcmr_abstract <- get_info_from_page(page, # scrape the "abstract"
                                    "#Abs1-content p:nth-child(2)")
  jcmr_method <- get_info_from_page(page, # scrape the "method"
                                   "#Abs1-content p:nth-child(4)")
  jcmr_author <- get_info_from_page(page, # scrape the "author"
                                   "#main-content header")
  jcmr_type <- get_info_from_page(page, # scrape the "article"
                                  ".c-article-title+ .c-article-identifiers .c-article-ident:1")
  jcmr_date <- get_info_from_page(page, # scrape the "publish"
                                  ".c-article-identifiers__item time")
  tibble(date = jcmr_date, # create a tibble co
         title = jcmr_title,
         volume = jcmr_vol,
         authors = jcmr_author,
         type = jcmr_type,
         abstract = jcmr_abstract,
         method = jcmr_method)
}

```

KEYWORD 1: "Machine Learning" 3 PAGES OF WEBSITE

```

# Scraping links from PAGE 1 - MACHINE LEARNING
pg1_ml <- scrape_list_links1("https://link.springer.com/search?query=machine+learning&content=Machine+Learning")

# Scraping links from PAGE 2 - MACHINE LEARNING
pg2_ml <- scrape_list_links1("https://link.springer.com/search?query=machine+learning&content=Machine+Learning")

# Scraping links from PAGE 3 - MACHINE LEARNING

```

```
pg3_ml <- scrape_list_links1("https://link.springer.com/search?query=machine+learning&content-t")

# Combining two pages into one list of links (55 links total)
ml <- list(pg1_ml, pg2_ml, pg3_ml)
clean_ml <- unlist(ml)

# Scraping information from each individual article
details_kwl <- purrr::map(clean_ml, scrape_individual_article) # Using our pre-made f
bind_rows(details_kwl)
```

```
# A tibble: 22 x 7
  date           title           volume authors type abstract method
  <chr>          <chr>          <chr> <chr>  <chr> <chr>  <chr>
1 10 March 2022 Precision measurement ~ Volum~ "Rhodr~ Rese~ Measure~ A ful~
2 27 February 2023 A deep learning approa~ Volum~ "Sachi~ Tech~ Cardiac~ Cardi~
3 27 June 2022 Radiomics and deep lea~ Volum~ "Ahmed~ Rese~ Myocard~ We ev~
4 20 June 2022 Highlights of the Virt~ Volum~ "Vinee~ Revi~ <NA> <NA>
5 07 December 2020 Deep learning to diagn~ Volum~ "Nicol~ Rese~ Cardiov~ 1.5 T~
6 07 October 2019 Machine learning in ca~ Volum~ "Tim L~ Revi~ <NA> <NA>
7 07 November 2022 Automatic segmentation~ Volum~ "Javie~ Rese~ Computa~ 90 CH~
8 06 January 2022 Accelerated cardiac T1~ Volum~ "Rui G~ Rese~ To deve~ We im~
9 11 March 2021 Fully-automated global~ Volum~ "Sona ~ Rese~ Cardiov~ Convo~
10 07 January 2019 Machine learning deriv~ Volum~ "Alex ~ Rese~ Phase c~ A mac~
# i 12 more rows
```

```
# Turning the binded rows into a tibble
details_kwl <- bind_rows(details_kwl)
```

KEYWORD 2: “Deep Learning” 3 PAGES OF WEBSITE

```
# Scraping links from PAGE 1 - DEEP LEARNING
pg1_dl <- scrape_list_links1("https://link.springer.com/search?query=deep+learning&content-t")

# Scraping links from PAGE 2 - DEEP LEARNING
pg2_dl <- scrape_list_links1("https://link.springer.com/search?query=deep+learning&content-t")

# Scraping links from PAGE 3 - DEEP LEARNING
pg3_dl <- scrape_list_links1("https://link.springer.com/search?query=deep+learning&content-t")

# Combining two pages into one list of links (54 links total)
```

```

dl <- list(pg1_dl, pg2_dl, pg3_dl)
clean_dl <- unlist(dl)

# Scraping information from each individual article
details_kw2 <- purrr::map(clean_dl, scrape_individual_article) # Using our pre-made f
bind_rows(details_kw2)

```

```

# A tibble: 3 x 7
  date   title volume authors type  abstract method
  <chr> <chr> <chr>  <chr> <chr> <chr>    <chr>
1 <NA>  <NA>  <NA>   <NA>  <NA>  <NA>    <NA>
2 <NA>  <NA>  <NA>   <NA>  <NA>  <NA>    <NA>
3 <NA>  <NA>  <NA>   <NA>  <NA>  <NA>    <NA>

```

```

# Turning the binded rows into a tibble
details_kw2 <- bind_rows(details_kw2)

```

KEYWORD 3: “Convolution” 2 PAGES OF WEBSITE

```

# Scraping links from PAGE 1 - CONVOLUTION
pg1_con <- scrape_list_links1("https://link.springer.com/search?query=convolution&content-ty

# Scraping links from PAGE 2 - CONVOLUTION
pg2_con <- scrape_list_links1("https://link.springer.com/search?query=convolution&content-ty

# Combining two pages into one list of links (39 links total)
con <- list(pg1_con, pg2_con)
clean_con <- unlist(con)

# Scraping information from each individual article
details_kw3 <- purrr::map(clean_con, scrape_individual_article) # Using our pre-made f
bind_rows(details_kw3)

```

```

# A tibble: 2 x 7
  date   title volume authors type  abstract method
  <chr> <chr> <chr>  <chr> <chr> <chr>    <chr>
1 <NA>  <NA>  <NA>   <NA>  <NA>  <NA>    <NA>
2 <NA>  <NA>  <NA>   <NA>  <NA>  <NA>    <NA>

```

```
# Turning the binded rows into a tibble
details_kw3 <- bind_rows(details_kw3)
```

KEYWORD 4: "Artificial Intelligence" 2 PAGES OF WEBSITE

```
# Scraping links from PAGE 1 - ARTIFICIAL INTELLIGENCE
pg1_ai <- scrape_list_links1("https://link.springer.com/search?query=artificial+intelligence")

# Scraping links from PAGE 2 - ARTIFICIAL INTELLIGENCE
pg2_ai <- scrape_list_links1("https://link.springer.com/search?query=artificial+intelligence")

# Combining two pages into one list of links (38 links total)
ai <- list(pg1_ai, pg2_ai)
clean_ai <- unlist(ai)

# Scraping information from each individual article.
details_kw5 <- purrr::map(clean_ai, scrape_individual_article) # Using our pre-made function
bind_rows(details_kw5)
```

```
# A tibble: 2 x 7
  date   title volume authors type abstract method
  <chr> <chr> <chr>   <chr>   <chr> <chr>   <chr>
1 <NA>  <NA>   <NA>    <NA>    <NA> <NA>    <NA>
2 <NA>  <NA>   <NA>    <NA>    <NA> <NA>    <NA>
```

```
# Turning the binded rows into a tibble
details_kw5 <- bind_rows(details_kw5)
```

```
# Combining all data tables with different key words into one tibble
all_articles <- bind_rows(
  details_kw1 |>
    mutate(keyword = "machine learning"),
  details_kw2 |>
    mutate(keyword = "deep learning"),
  details_kw3 |>
    mutate(keyword = "convolution"),
  details_kw5 |>
    mutate(keyword = "artificial intelligence")
)
```

```
# Creating the CSV file  
write_csv(all_articles, "springer_articles.csv")
```